

EcoExplain-VLM

Complete Research Learning and Implementation Roadmap

Evaluating Evidence-Grounded Explanations from Vision-Language Models for Forest Disturbance Assessment

Conference 3 — Topic 9: Explainable AI for Ecological Inference and Policy Support

Prepared as a 40-day independent research plan

1. Final Project Definition

Project title

EcoExplain-VLM: Evaluating Evidence-Grounded Explanations from Vision-Language Models for Forest Disturbance Assessment

Central research question

Can vision-language models correctly classify forest vegetation change and generate explanations that are supported by multimodal remote-sensing and climate evidence?

Project framing

The paper is primarily an evaluation and explainability benchmark, not a new remote-sensing model. It evaluates multimodal evidence use, explanation groundedness, uncertainty and confidence calibration.

2. Knowledge to Build Before Implementation

2.1 Remote-Sensing Foundations

Spatial concepts

- Raster data
- Pixels and bands
- Spatial, temporal and spectral resolution
- Coordinate reference systems
- Region of interest
- Image compositing
- Resampling
- Patch extraction
- Spatial autocorrelation
- Spatial data leakage

Optical remote sensing

- Surface reflectance and atmospheric correction
- Cloud and cloud-shadow masking
- Natural-colour and false-colour composites
- Red, green, blue and near-infrared bands
- Vegetation spectral response
- Seasonal and phenological variation

Required Sentinel-2 bands

- B2: Blue

- B3: Green
- B4: Red
- B8: Near infrared

Vegetation indices

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

- Healthy vegetation generally has higher NIR reflectance
- NDVI can saturate in dense forests
- Soil, atmosphere, cloud and season affect NDVI
- NDVI decline does not automatically prove forest degradation

$$\text{EVI} = 2.5 \times (\text{NIR} - \text{Red}) / (\text{NIR} + 6 \times \text{Red} - 7.5 \times \text{Blue} + 1)$$

- EVI requires the blue band
- EVI reduces some atmospheric and soil-background effects
- EVI may remain informative where NDVI saturates

SAR remote sensing

- Synthetic Aperture Radar and radar backscatter
- Decibel units
- VV and VH polarization
- Surface roughness, vegetation structure and moisture
- Speckle
- Ascending and descending orbits
- Incidence angle
- Cloud-independent observation

Caution: VV or VH change is not definitive vegetation loss. SAR also responds to moisture, acquisition geometry and surface structure.

2.2 Climate-Data Foundations

- Weather versus climate
- Temperature and rainfall anomalies
- Monthly climatology
- Standardized anomaly
- Drought
- Lagged climate effects
- Spatial-resolution mismatch
- Ecological confounding

Monthly anomaly = observed value – historical mean for the same month

ERA5-Land is much coarser than Sentinel imagery. Climate variables must be described as regional context, not 10 m patch measurements.

2.3 Forest-Disturbance Concepts

- Forest and canopy cover
- Vegetation condition
- Disturbance
- Degradation versus deforestation
- Fire disturbance
- Vegetation decline and increase
- Recovery and resilience
- Phenology and seasonal variation
- Drought stress
- Anthropogenic and natural disturbance

Required conservative terminology

- Use 'vegetation decline signal' rather than automatically claiming forest degradation.
- Use 'fire-associated disturbance' only when independent fire data supports the claim.
- Use 'vegetation increase' rather than recovery unless a longer time series demonstrates recovery.
- Do not infer causation from correlation between climate anomalies and vegetation change.

3. Explainable AI Foundations

Term	Meaning
Interpretability	How easily a human can understand a system or its behaviour.
Explanation	A human-readable account of evidence supporting a conclusion.
Local explanation	An explanation for one patch assessment.
Global explanation	A summary of model behaviour across the benchmark.
Intrinsic explainability	A model or decision system that is understandable by design.
Post-hoc explanation	An explanation generated after a prediction.
Plausibility	Whether an explanation sounds reasonable to a person.
Groundedness	Whether explanation claims are supported by the supplied input.
Faithfulness	Whether the explanation reflects the process responsible for the output.
Hallucination	A claim not supported by the supplied evidence.
Contradiction	A statement that conflicts with the input.
Evidence omission	Failure to mention important supplied evidence.

Selective explanation	Using supporting evidence while ignoring conflicting evidence.
Counterfactual explanation	How a conclusion would change if an input changed.
Confidence calibration	Agreement between reported confidence and empirical correctness.

For black-box APIs, true internal faithfulness is difficult to establish. The study should claim evaluation of input groundedness, evidence consistency, contradiction, evidence sensitivity and explanation stability—not proof of internal reasoning.

4. Vision-Language Model Foundations

- Vision encoder
- Language model
- Multimodal projection
- Image-text alignment
- Instruction tuning
- Zero-shot and few-shot evaluation
- Prompt sensitivity
- Structured generation
- Decoding temperature
- Context window
- Multimodal hallucination
- Visual grounding
- Numerical reasoning
- Modality dominance

Model selection

1. One strong proprietary VLM
2. One additional accessible hosted VLM
3. One open-source VLM

Record the full identifier, checkpoint or API version, access date, temperature, image resolution, prompt, token settings and repetitions. Do not replace models during the final experiment.

5. Benchmark Design

Study area

Use one geographically compact forest area, such as a defined part of the Nilgiri region. Avoid the full Western Ghats because ecological and seasonal heterogeneity would increase confounding and workload.

Temporal design

Compare matched seasonal windows from two years. Use median composites rather than single images to reduce cloud and acquisition noise.

Sample design

- Target 300–500 patch pairs
- Start with a 40–60 sample pilot
- Use approximately balanced operational classes
- Use 64×64 Sentinel pixels per patch, approximately 640 m × 640 m
- Use spatially separated evaluation partitions

Reference categories

4. Stable vegetation
5. Vegetation decline
6. Vegetation increase
7. Fire-associated disturbance
8. Uncertain or conflicting evidence

Standard VLM evidence card

- Before and after natural-colour images
- Before and after false-colour images
- NDVI before, after and change
- EVI before, after and change
- VV before, after and change
- VH before, after and change
- Temperature anomaly
- Rainfall anomaly
- Optional fire indicator
- Acquisition season and data-quality information

Do not reveal coordinates or location names in the primary evaluation; otherwise, a model may use memorized geographic information.

6. Ground-Truth Strategy

Stage 1: Candidate generation

Use change distributions to identify stable, declining and increasing candidates. Begin with region-specific quantiles rather than claiming universal ecological thresholds.

Stage 2: Quality filtering

- Exclude: Excessive cloud or cloud shadows
- Exclude: Missing SAR observations
- Exclude: Water-dominated patches
- Exclude: Severe spatial misalignment
- Exclude: Season mismatch
- Exclude: Insufficient forest cover
- Exclude: Image artefacts

Stage 3: Manual validation

Inspect before/after imagery, vegetation-index maps, SAR observations, fire products and longer temporal trends where needed.

Stage 4: Independent audit

Have a second reviewer inspect a subset. If possible, obtain a remote-sensing or ecology audit for 30–50 samples. Describe labels as operational benchmark labels, not perfect ecological truth.

7. Experimental Conditions

Condition A — Images only

Tests visual interpretation.

Condition B — Indicators only

Tests numerical and tabular reasoning.

Condition C — Combined evidence

Tests multimodal evidence integration.

Optional Condition D — Structured explanation

Requires supporting evidence, conflicting evidence, conclusion and uncertainty separately.

Every selected patch should be evaluated by every model under every retained condition.

8. Structured Output

```
{ "change_class": "stable", "disturbance_present": false, "supporting_evidence": ["NDVI remained stable", "EVI remained stable"], "conflicting_evidence": ["VH decreased slightly"], "explanation": "Optical indicators remained stable; the small VH reduction alone is insufficient to establish disturbance.", "confidence": 0.76, "insufficient_evidence": false }
```

Request a short evidence-based rationale, not hidden chain-of-thought.

9. Pre-Registered Hypotheses

- H1: Combined images and indicators produce higher classification performance than images alone.
- H2: Combined input improves evidence coverage but may increase unsupported causal claims.
- H3: Models differ in classification, groundedness and confidence calibration.
- H4: Higher reported confidence is associated with correctness, but models are overconfident.
- H5: Structured explanations contain fewer contradictions than unrestricted explanations.

10. Evaluation Metrics

Classification

- Accuracy
- Balanced accuracy
- Macro precision and recall
- Macro-F1 as the primary classification metric
- Per-class F1
- Confusion matrix

Evidence and explanation

- Evidence precision, recall and F1
- Unsupported-evidence rate
- Contradiction rate
- Important-evidence omission
- Evidence correctness
- Evidence completeness
- Uncertainty awareness
- Causal restraint
- Overall usefulness

Confidence and consistency

- Brier score
- Expected Calibration Error
- Reliability diagrams
- Accuracy by confidence interval
- Class agreement over repeated runs
- Confidence variance
- Evidence overlap

- Explanation consistency

11. Statistical Analysis

The design is repeated-measures: identical patches are evaluated across models and input conditions.

- Bootstrap confidence intervals sampled by patch
- Paired model comparisons
- Mixed-effects logistic regression for correctness
- Ordinal mixed-effects models for human ratings
- Model and input condition as fixed effects
- Patch as a random effect
- Holm correction for multiple comparisons
- Effect sizes alongside p-values
- Weighted Cohen's kappa for two raters
- Krippendorff's alpha for multiple raters or missing ratings
- Adjudication only after independent ratings are saved

12. Forty-Day Implementation Plan

Period	Phase	Required output
Days 1–4	Foundations and scope	Learn core concepts; freeze project definition, region candidates, claims and non-claims.
Days 5–7	Literature review	Review 20–30 papers; build literature matrix, research gap and related-work outline.
Days 8–10	Protocol design	Freeze sample, classes, exclusions, conditions, hypotheses, metrics and analysis plan.
Days 11–15	Pipeline prototype	Build Earth Engine processing and export 40–60 pilot samples.
Days 16–18	Pilot validation	Inspect all samples, verify calculations, prompt structure and JSON parsing.
Days 19–22	Full benchmark	Create and validate 300–500 samples, spatial splits and reference evidence.
Days 23–25	Prompt finalisation	Use development data, freeze prompt v1.0 and experiment configuration.
Days 26–29	Model inference	Run models; preserve raw outputs, metadata, failures, usage and cost.
Days 30–32	Automatic evaluation	Compute classification, evidence, contradiction, calibration and consistency results.
Days 33–35	Human evaluation	Blind and randomize 80–100 explanations; obtain

		independent ratings.
Days 36–37	Statistical analysis	Run bootstrap, mixed-effects, corrected comparisons and sensitivity analysis.
Days 38–40	Paper and release	Complete paper, code, prompts, metadata, data card and acquisition instructions.

13. Recommended Repository Structure

ecoexplain-vlm/ ├── README.md ├── configs/ ├── data/ | ├── raw_metadata/ | ├── processed/ | ├── labels/ | ├── splits/ ├── earth_engine/ ├── src/ | ├── build_cards.py | ├── query_models.py | ├── parse_responses.py | ├── evaluate_predictions.py | ├── evaluate_evidence.py | ├── calibration.py ├── prompts/ ├── annotations/ ├── notebooks/ ├── results/ ├── tests/ └── paper/

14. Main Risks and Controls

Risk	Control
Subjective ground truth	Operational rules, manual validation and independent audit.
Season mistaken for disturbance	Compare matched seasonal windows.
Cloud artefacts	Cloud/shadow masking and manual inspection.
SAR misinterpretation	Use SAR as supporting rather than definitive evidence.
Climate resolution mismatch	Use regional contextual anomalies.
Convincing unsupported explanations	Measure grounding, contradiction and causal restraint.
Prompt overfitting	Separate development and untouched test samples.
Spatial leakage	Use geographic splitting.
API model changes	Record exact model identifiers and dates.
No ecology expert	Limit claims and seek a small external audit.
Excess workload	Start with pilot; make fire optional; cap final sample count if needed.

15. Knowledge Readiness Checklist

- ☐ What is the difference between surface reflectance and an RGB image?
- ☐ Why does vegetation reflect strongly in near infrared?
- ☐ How are NDVI and EVI calculated?
- ☐ Why can NDVI decline without forest disturbance?
- ☐ What do VV and VH approximately represent?
- ☐ Why can SAR change because of rainfall or moisture?

- ☐ Why must seasonal observation windows be matched?
- ☐ Why is ERA5 not a 10 m measurement?
- ☐ What is the difference between plausible and grounded explanations?
- ☐ Why can black-box rationale text not prove internal faithfulness?
- ☐ How will reference labels be generated independently from VLM outputs?
- ☐ What constitutes an unsupported ecological claim?
- ☐ What is the experimental unit?
- ☐ Why must results be compared by patch?
- ☐ What is the primary evaluation metric?

16. Expected Paper Contribution

A reproducible multimodal benchmark and controlled evaluation of VLM forest-change classification, input-grounded explanation, multimodal evidence use, contradiction, stability and confidence calibration.

Claims to avoid

- Automatic diagnosis of complete forest health
- Causal identification of disturbance drivers from correlation
- True internal explanation faithfulness for black-box models
- 10 m climate measurements from ERA5-Land
- Generalization to all forest ecosystems from one study region

17. Immediate Next Step

Before downloading the full dataset, prepare a one-page protocol fixing the study boundary, observation periods, sample unit, class definitions, exclusion rules, hypotheses, models, metrics and analysis plan. Then implement only the 40–60-sample pilot. Scale to the complete benchmark only after the pilot passes data-quality and prompt-validation checks.